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FIT3080 – Artificial Intelligence

Intelligent Agents Chapter 2

Outline

- **Agents and environments**
- **Rationality**
- **PEAS (Performance measure, Environment, Actuators, Sensors)**
- **Environment types**
- **Agent types**

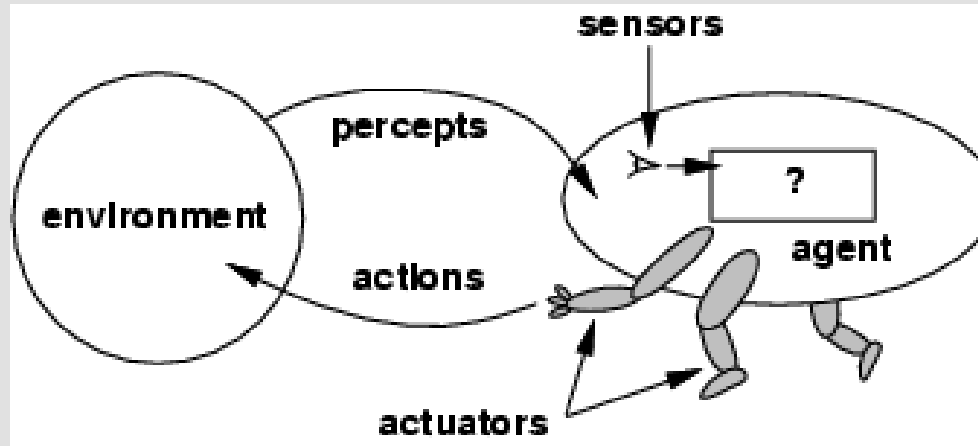


Agents

- An **agent** is anything that can be viewed as **perceiving** its **environment** through **sensors**, and **acting** upon that environment through **actuators**
- **Human agent:**
 - eyes, ears and other organs for sensors
 - hands, legs, mouth and other body parts for actuators
- **Robotic agent:**
 - cameras and infrared range finders for sensors
 - various motors for actuators
- **Software / virtual agent:**
 - keyboard input, file input, receiving from network
 - screen, file output, sending to network

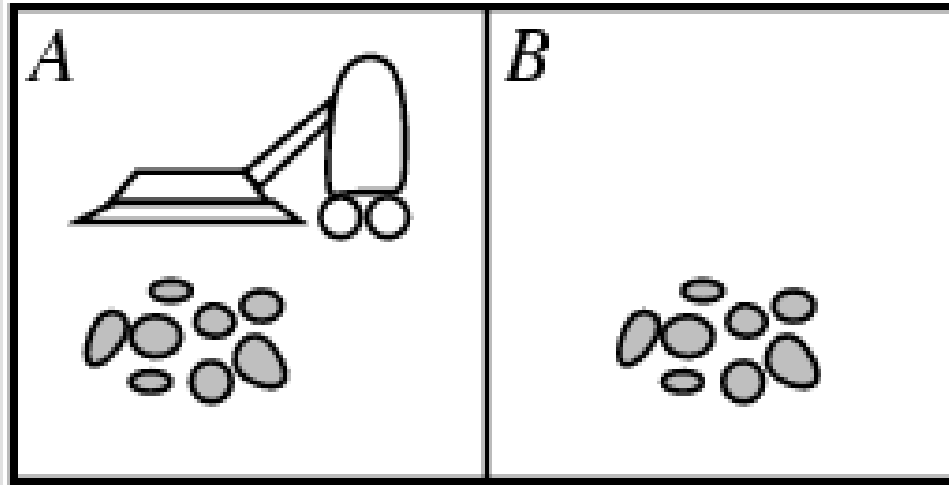


Agents and Environments



- The **agent function** maps from percept histories to actions: $f: \mathcal{P}^* \rightarrow \mathcal{A}$
- The **agent program** runs on the physical **architecture** to produce f
- **agent = architecture + program**

Example: Vacuum-cleaner World and Agent



- **Percepts:** location and contents, e.g., [A,Dirty]
- **Actions:** *Left, Right, Suck*
- **Program:**
 If *status=Dirty* **return** Suck
 Elseif *Location=A* **return** Right
 Elseif *Location=B* **return** Left

**Is this a
rational agent?**



Rationality and Rational Agents

- **Rationality depends on**
 - The set of actions available to the agent
 - The agent's prior knowledge of the environment
 - Additional information, gathered or inferred from percepts
 - Performance measure
- **Definition:**

*For each possible **percept sequence**, a rational agent should select an **action** that is expected to maximize its **performance measure**, given the evidence provided by the **percept sequence** and the agent's **built-in knowledge***

Rational, Autonomous Agents

- Rationality is NOT omniscience
- Agents can perform actions to modify future percepts in order to obtain useful information
→ **exploration, learning**
- An agent is **autonomous** if its behavior is determined by its own experience



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Task Environment

Task Environment – PEAS

To design a rational agent, we must specify the Task Environment

- **PEAS**
 - Performance measure
 - Environment
 - Actuators
 - Sensors



PEAS – Automated Taxi Example (I)



Image credits: Total Recall (1990)



PEAS – Automated Taxi Example (II)

- **Performance measure**
 - Safe, fast, legal, comfortable trip, minimize fuel consumption, maximize profit
- **Environment**
 - Road types, road contents, customers, operating conditions
- **Actuators**
 - Control over the car, interfaces for informing other vehicles and informing passengers
- **Sensors**
 - Cameras, sonar, speedometer, GPS, odometer, engine sensors, interface for receiving information from other vehicles and passengers (e.g., speech recognizer)

PEAS – Automated Taxi Example (III)

Performance measure considerations

- Rational agent: maximize expected performance measure, considering both now and the future

Performance	Score	
Safe	Every crash	-100
	Arrives unharmed	+500
Fast	Every second	-0.1
Legal	Every violation	-500
Comfortable	Every speed bump	-10
	Sudden turn, braking	-50
Fuel consumption	Every litre	-0.5

Design a performance measure according to what you want

PEAS – Internet Shopping Agent Example

- **Performance measure**
 - cheap, good quality, appropriate product
- **Environment**
 - current WWW sites, vendors
- **Actuators**
 - Packages that display to user, follow URL, fill in forms
- **Sensors**
 - Packages that read HTML pages (text, graphics, scripts)



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Environment Types

Environment Types (I)

Environment type largely determines agent design

Fully or partially observable



Fully (partially) observable –
An agent's sensors give it access to the complete state of the environment at all times

Known or unknown



Known (unknown) – An agent knows the “laws” of the environment

Environment Types (II)

Environment type largely determines agent design

Single or multi agent



Single (multi) agent – An agent operating by itself in an environment

Deterministic or stochastic



Deterministic (stochastic) – The next state is completely determined by the current state and the action executed by the agent

Environment Types (III)

Environment type largely determines agent design

Episodic or sequential



Episodic (sequential) – The agent's experience is divided into atomic *episodes*. The next episode does NOT depend on previous actions

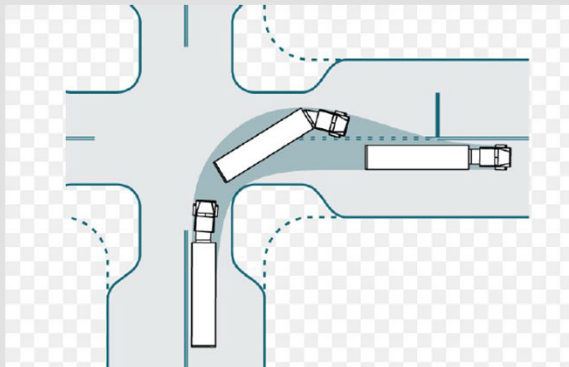
- In each episode an agent perceives a percept and performs a single action

Environment Types (IV)

Environment type largely determines agent design



Static (dynamic) – The environment is unchanged while an agent is deliberating



Discrete (continuous) – Pertains to the state of the environment, the way time is handled, and the percepts and actions of the agent

Environment Types – Examples

	8-puzzle	Back-gammon	Medical diagnosis	Taxi
Observable?				
Known?				
Single agent?				
Deterministic?				
Episodic?				
Static?				
Discrete?				

The real world is partially observable, unknown, multi-agent, stochastic, sequential, dynamic, continuous

Environments and Methodologies

	Search	Logical inference	Bayesian networks	Markov decision processes	Reinforcement learning	Machine learning
Observable?	✓	✓		✓	✓	
Known?	✓	✓	✓	✓	✗	✗
Single agent?						
Deterministic?	✓	✓	✗	✗	✗	
Episodic?						
Static?	(✓)	✓	✓	✓	✓	✓
Discrete?	(✓)	✓	(✓)	✓	✓	



Agent Functions and Programs

- An agent is completely specified by the agent function that maps percept sequences to actions
- Aim: design a program that implements the rational agent function concisely



Agent Types

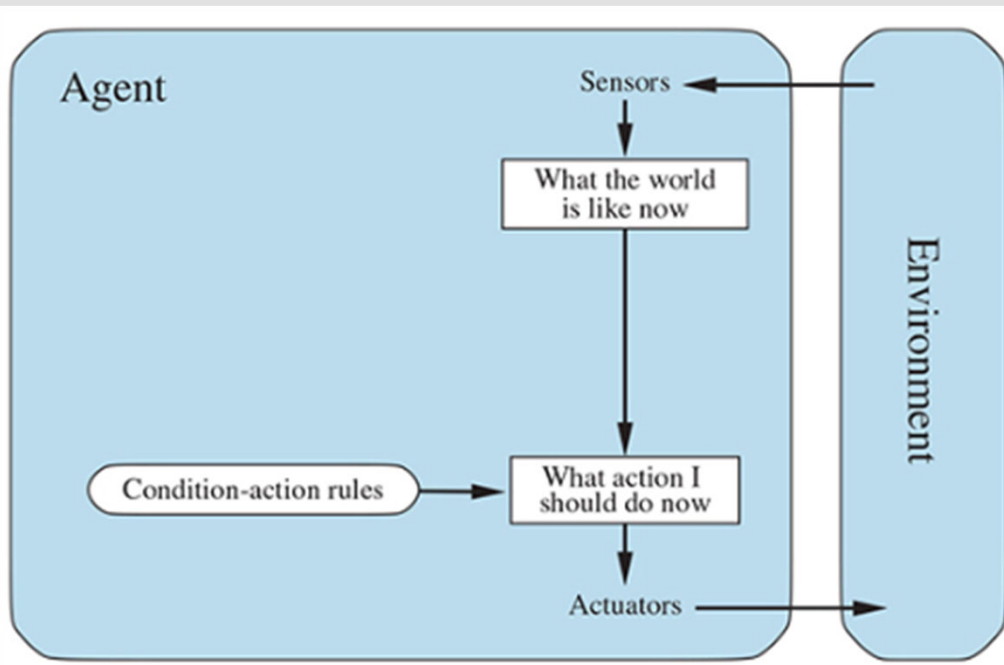
Based on the function = how actions are selected

Agent Type	Action selected based on
Simple reflex	current percepts
Model based	+ internal state (world model)
Goal based	Model-based + goal
Utility based	Model-based [+ goal] + utility function
Learning	performance element = above agent + critic + learning element + problem generator (exploratory)

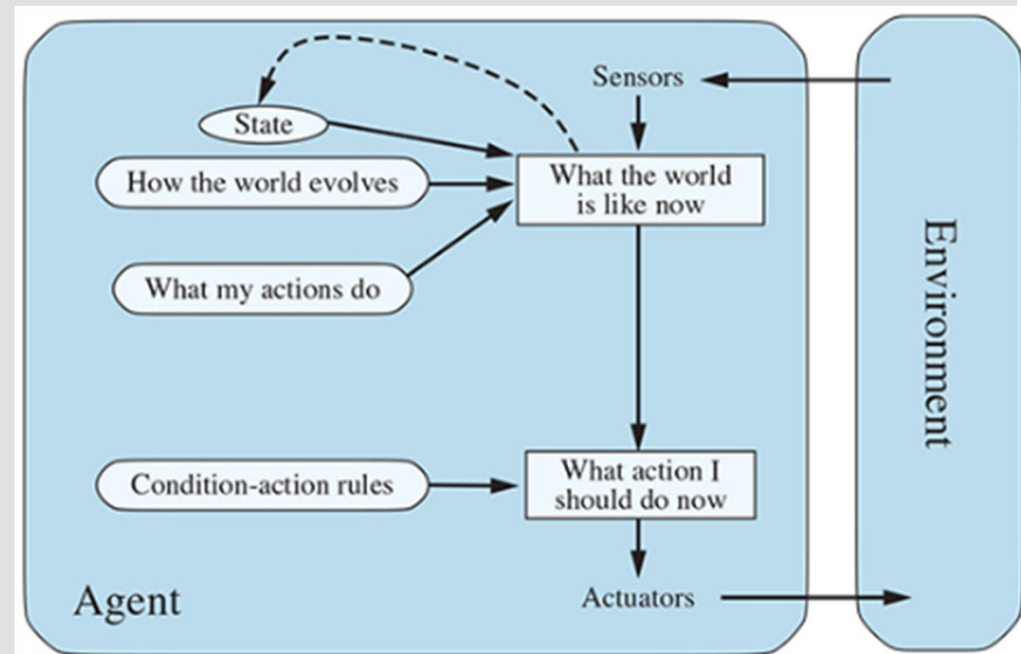
Agent Types – Taxi Example

Agent Type	Action
Simple reflex	brake when brake-lights of car in front light up
Model based	+ remember the roads travelled, time, state
Goal based	+ make a plan to reach a destination
Utility based	+ quickest with least petrol consumption
Learning performance elem +critic +learning element +problem generator	above agent observes the world & informs learning elem formulates new driving rules based on the feedback from critic + perf elem requirements might suggest some driving exercises

Reflex and Model-based Agent

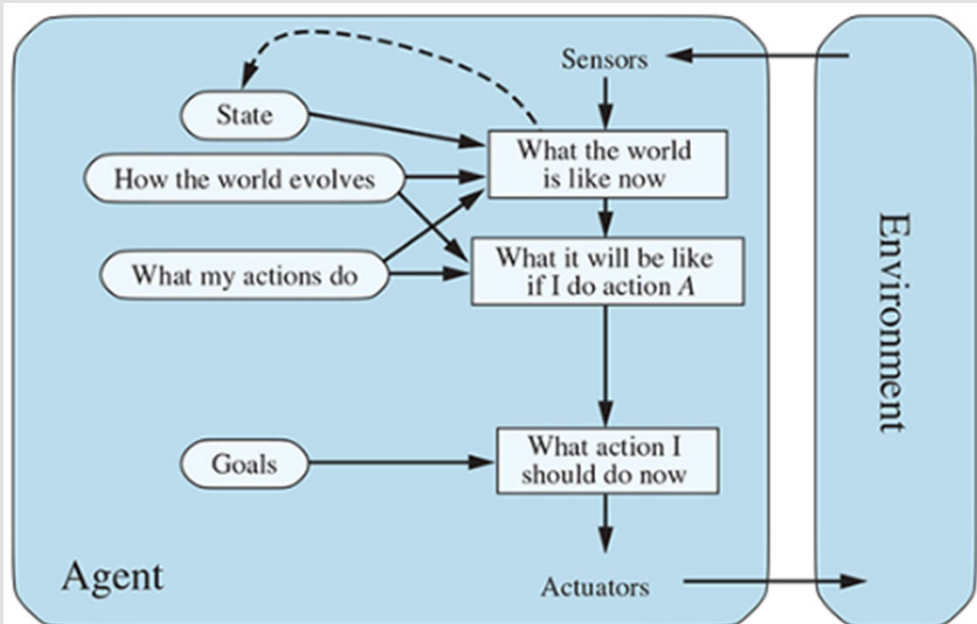


Reflex agent

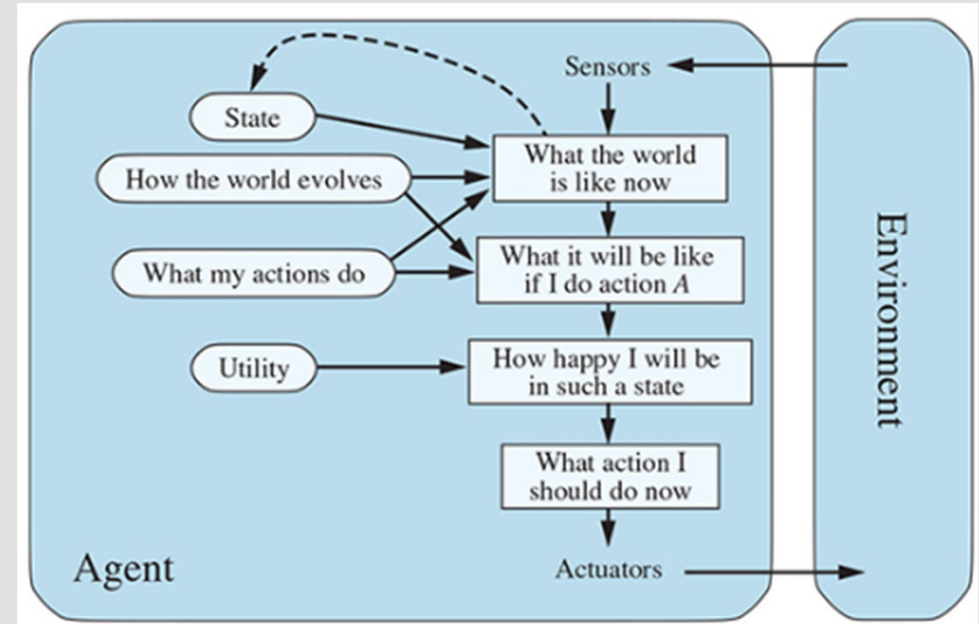


Model-based agent

Model-based (Goal- and Utility-based) Agent

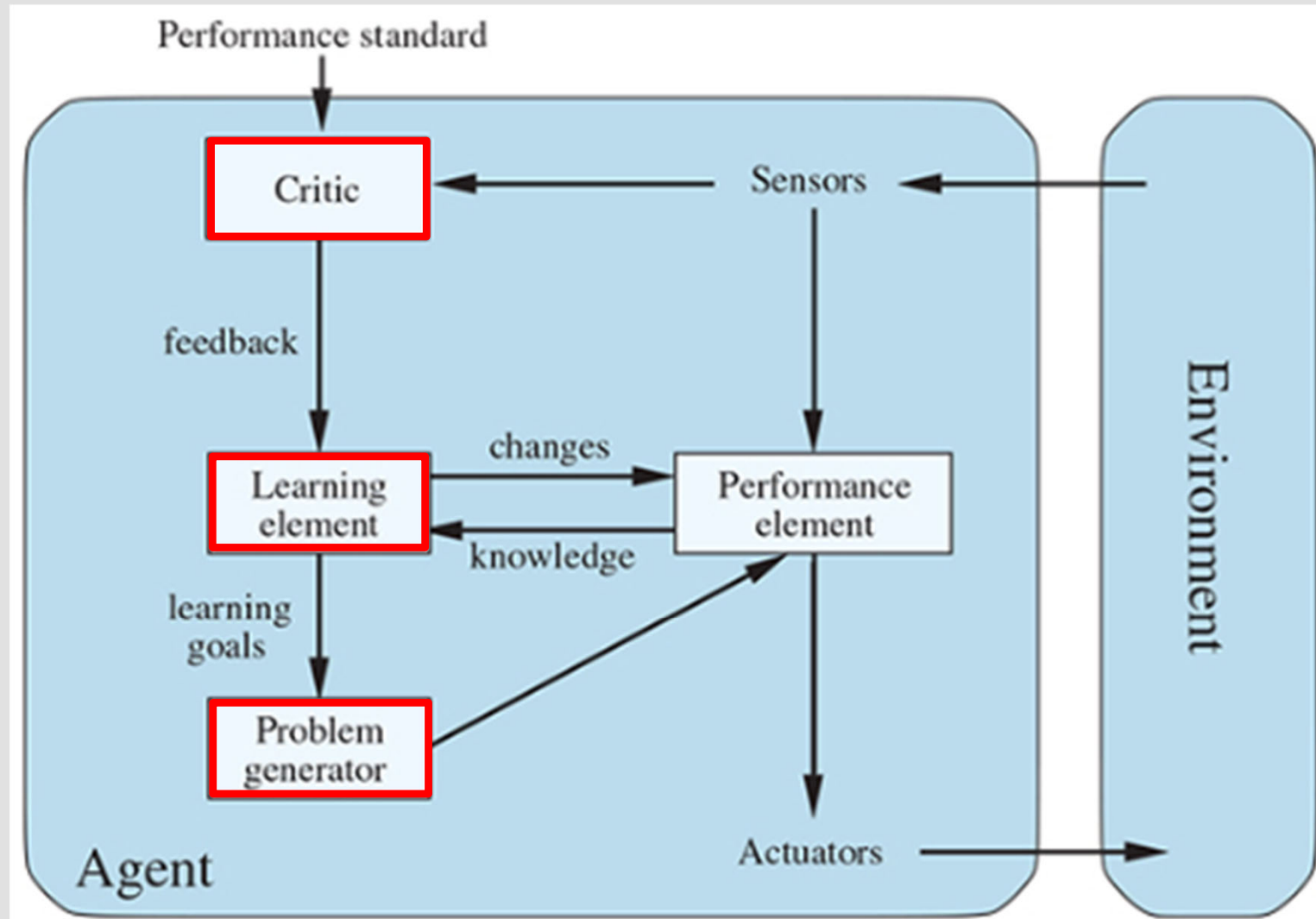


Model-based goal-based agent



Model-based utility-based agent

Learning Agent



How Components of Agent Programs Work?

Depends on the representations of states:

- **Atomic** – each state is indivisible (Search, Game playing, Markov Decision Processes)
- **Factored** – splits each state into attributes, each of which has a value (Propositional logic, Bayesian networks, Machine learning)
- **Structured** – represents how things are related to each other (First order logic, First-order probability models)

A more expressive representation can capture everything a less expressive representation can capture, but reasoning and learning is harder

Reading

- **Russell, S. and Norvig, P, *Artificial Intelligence – A Modern Approach* (4th ed), Prentice Hall, Chapter 2**



Next Lecture Topic

- **Lecture Topic 3**
 - Problem Solving as Search

